Appendix: Forecasting severe system pressure in the NHS

Description of development dataset

The development data is provided in a unified long format (features and outcome combined) with the following structure:

- **dt:** Date-time.
- **metric_name:** Name/description of metric.
- **value:** Measured value.
- **coverage:** Provider or geographic scope.
- **coverage_label:** Cleaned version of the coverage field.
- **variable_type**: Indicates if the metric is a feature or the outcome.

Metric time-ranges

The temporal coverage of each metric varies, as do their recording frequencies and calendar anchor points. These distinctions are summarised in figure 1.

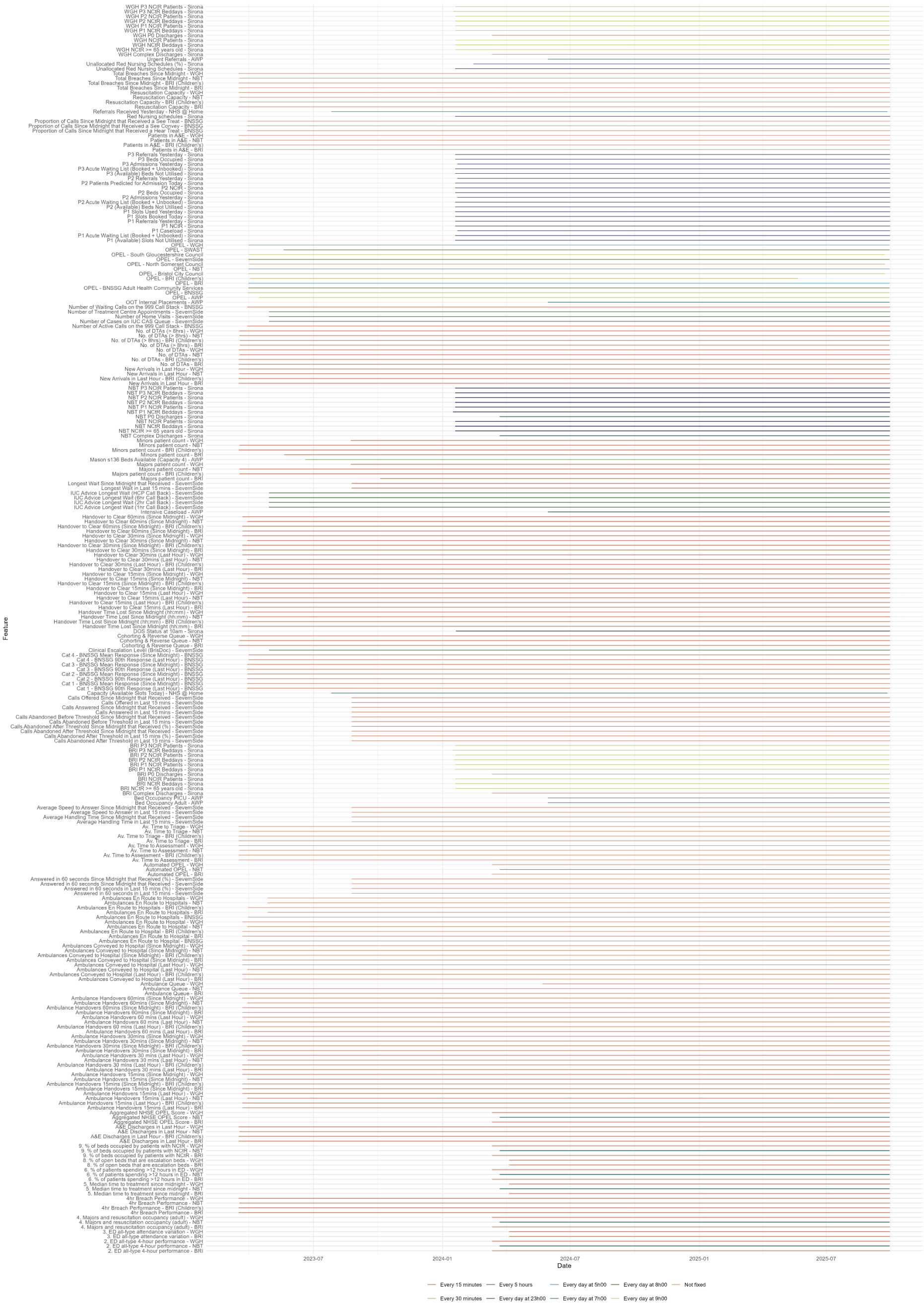

Figure 1: Temporal qualities of the explanatory feature set.

Outcome variable

Figure 2 visualises the Estimated avoidable deaths as a time series plot:

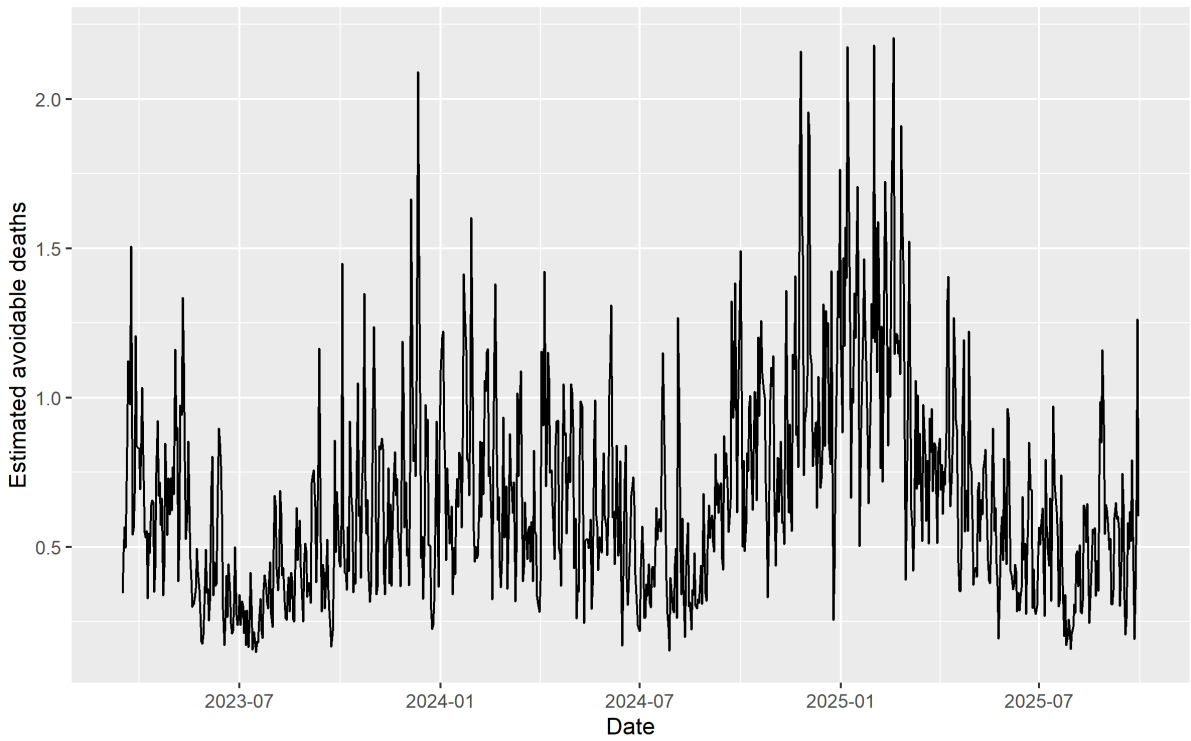

Figure 2: Time series of estimated avoidable deaths (outcome variable).

Glossary of terms

The metric names in the development dataset include various domain-specific terms and acronyms. Use the glossary below for definitions:

Term	Desciption
BNSSG	'Bristol, North Somerset & South Gloucestershire'.
SG	'South Gloucestershire'.
NS	'North Somerset'.
ICB	'Integrated Care Board'.
BRI	'Bristol Royal Infirmary'.
NBT	'North Bristol Trust' (Southmead Hospital).
WGH	'Weston General Hospital'.
NCtR	'No Criteria to Reside', a status given to patients who are medically fit to leave their current clinical setting but are otherwise blocked.

MIU	‘Minor Injury Unit’.
DtA	‘Decision to Admit', a status given to a patient in A&E who has had the decision that they require an admitted bed, these are counted as they often queue inside A&E.
D2A	‘Discharge to Assess', the name of the intermediate care service provided in BNSSG ICB.
P0, P1, P3, P4	Pathways 0-4 are different levels of intermediate care settings, into which patients are discharged from acute beds.
ED	‘Emergency Department'.
IUC	‘Integrated Urgent Care'.
HCP	‘Healthcare Professional'.
OPEL	‘Operational Pressures Escalation Levels', a composite metric indicating system pressure.
G&A	‘General & Acute’, a classification of bed type.
CAS	‘Clinical Assessment Service'.
Majors	Patients who are seriously ill or injured and require a high level of nursing care and medical intervention.
Minors	Patients with injuries or illnesses that are urgent but not life-threatening.
SBRU	‘South Bristol Community Hospital Rehabilitation Unit'
SWASFT	‘South Western Ambulance Service Foundation Trust'.
TOC	‘Transfer Of Care'.
C1, C2, C3, C4	Categories 1-4 of ambulance response (from most to least critical).
DOS	‘Directory Of Service'.
GP	‘General Practice'.
ICE	‘Inner City and East' locality.
OOT	‘Out Of Trust'.
PICU	‘Paediatric Intensive Care Unit.
SASS	‘Specialist and Support Services'